

Introduction

Recursion in java is a process in which a method calls itself continuously. A method in java that calls itself is called recursive method.

Syntax



```
returntype methodname( ){  
    //code to be executed  
    methodname( ); //calling same method  
}
```

Example 1: Infinite times



```
public class RecursionExample1 {  
    static void p(){  
        System.out.println("hello");  
        p();  
    }  
    public static void main(String[] args) {  
        p();  
    }  
}  
//it will print hello infinite times
```

Example 2: Finite times



```
public class RecursionExample2 {  
    static int count=0;  
    static void p(){  
        count++;  
        if(count<=5){  
            System.out.println("hello "+count);  
            p();  
        }  
    }  
    public static void main(String[] args) {  
        p();  
    }  
}  
  
//it will print hello 5 times
```

Example 3: Factorial Number



```
public class RecursionExample3 {  
    static int factorial(int n){  
        if (n == 1)  
            return 1;  
        else  
            return(n * factorial(n-1));  
    }  
    public static void main(String[] args) {  
        System.out.println("Factorial of 5 is: "+factorial(5));  
    }  
}
```